REFERENCE · CHAPTER 13

End-to-End Audit Attribution (W3C Trace, BigQuery Sink) & Kill Switch Runbook

Linking every SQL query to the human caller and agent turn, enforcing Principal Access Boundaries (PAB), and executing the 3-step emergency kill switch.

1. Closing the Audit Blind Spot (`T8`)

By default, Cloud SQL, AlloyDB, and Spanner disable `DATA_READ` and `DATA_WRITE` audit logs, and logs in `_Default` expire after 30 days. Our control plane enables Data Access logs across all data stores, routes them to a 400-day partitioned BigQuery sink, propagates W3C `traceparent` from the client edge into SQL comments/job labels, and streams full agent trajectories via ADK's `BigQueryAgentAnalyticsPlugin` .

2. ADK `BigQueryAgentAnalyticsPlugin` & `BigQueryLoggerConfig`

Attached to the ADK `App` container (or enabled at scaffold time via `uvx agent-starter-pack create . --bq-analytics`), `BigQueryAgentAnalyticsPlugin` streams structured agent lifecycle events directly into BigQuery using the **BigQuery Storage Write API**, supporting auto-schema upgrades, GCS offloading for multimodal content, and OpenTelemetry `trace_id` / `span_id` correlation:

PARAMETER / FIELD	TYPE / SCOPE	SECURITY & GOVERNANCE PURPOSE
`trace_id` / `span_id`	Event Column	OpenTelemetry distributed trace context; joins directly with `sqlcommenter` `traceparent` tags in 400-day Database Data Access Audit Logs.
`tool_provenance`	Event Metadata	Identifies tool execution origin: `LOCAL` , `MCP` , `SUB_AGENT` , `A2A` , or `TRANSFER_AGENT` .
`content_formatter`	`BigQueryLoggerConfig`	Pre-ingestion Python callback that scrubs or redacts residual SSNs, credit cards, or bearer tokens before events leave the container (`T2` / `LLM02`).

PARAMETER / FIELD	TYPE / SCOPE	SECURITY & GOVERNANCE PURPOSE
`event_allowlist`	`BigQueryLoggerConfig`	Optional filter restricting logged events (e.g., `USER_MESSAGE_RECEIVED`, `LLM_REQUEST`, `LLM_RESPONSE`, `TOOL_STARTING`, `TOOL_COMPLETED`, `TOOL_ERROR`).
`max_content_length` & `shutdown_timeout`	`BigQueryLoggerConfig`	Caps inline string payloads (with optional GCS bucket offloading for large/multimodal content) and flushes buffered events on container shutdown.

3. The 3-Step Emergency Kill Switch Runbook

STEP	ACTION	COMMAND / CONTROL PLANE TARGET	FORENSIC IMPACT
Step 1: Block Agent Principal	Apply an immediate IAM Deny policy & Principal Access Boundary (PAB) ceiling on the agent's SPIFFE identity, and remove the agent's egress permission at the Egress Gateway (`iap.googleapis.com/resources.egressViaIAP` under Unified Access Policies, or `roles/iap.egressor` under the earlier IAP policy version).	`gcloud iam deny-policies create` / `IAM PAB`	Halts all tool and database calls within seconds while keeping the container intact.
Step 2: Revoke Delegated Tokens	Revoke the user's sessions and refresh tokens at the identity provider, remove any credentials vaulted in Agent Identity auth manager, and rotate the `ALLOWED_KIDS` KMS context signing key version if needed.	Identity provider / auth manager / `Cloud KMS`	Invalidates any in-flight tokens immediately across all MCP servers.
Step 3: Preserve Evidence	Snapshot Cloud Run / Agent Runtime revisions, export `Memory Bank` sessions, and query the 400-day BigQuery audit sink by `trace_id` and `spiffe_id`.	`BigQuery Audit Sink` + `Security Command Center`	Preserves complete chain of custody for incident response.